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SCANNING LINEAR ACCELERATOR SYSTEM FOR PRODUCING X-RAYS AT CONTROLLED DOSE AND REPETITION RATES

Abstract

A linac-based X-ray system for cargo scanning and imaging applications uses linac design, RF power control, beam current control, and beam current pulse duration control to provide stable sequences of pulses having different energy levels or different doses.

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Background/Summary

RELATED APPLICATION [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 15/484,065 filed Apr. 10, 2017, which in turn is a continuation-in-part of U.S. patent application Ser. No. 14/192,864, filed on Feb. 27, 2014, now U.S. Pat. No. 9,622,333, as well as U.S. patent application Ser. No. 14/634,361, filed Feb. 27, 2015, now U.S. Pat. No. 9,661,734, and claims the benefit of each of the foregoing, all of which are incorporated herein by reference for all purposes.

FIELD OF THE INVENTION

[0002] The present invention relates generally to X-ray devices, and more particularly relates to linac-based scanning systems and methods able to generate pulses of different energies or doses or both.

BACKGROUND OF THE INVENTION

[0003] Linear accelerator (sometimes “linac” hereinafter) X-ray generating systems have been in use in the medical environment for a number of years. More recently, such systems have begun to be used in the industrial environment, particular for cargo scanning. To distinguish the materials inside a cargo container, X-ray pulses of different energies have been used. Energies in the range of 2 MeV to 10 or more MeV have been proposed in the literature, and commercial devices offering energies at approximately 4 MeV and 6 MeV are commercially available. In many instances, both medical and industrial linac X-ray sources are excited by RF sources operating at or near the S-band, roughly 2.998 GHz. In each case, the linac accelerates a stream of electrons in conjunction with RF excitation, and the linac must be designed such that its resonance can be matched by the frequency of the RF source. Once the electrons have been sufficiently accelerated, if X-rays are desired, they strike a target, such as tungsten, resulting in the emission of high energy X-rays that can be used for medical treatment, materials processing, scanning of cargo, and other applications. These mega-electron-volt (MeV) X-ray applications have provided significant benefits in many fields.

[0004] The most prevalent medical and industrial linacs are resonant structures, and require an RF excitation source, typically either a pulsed magnetron or a pulsed klystron. To couple the power from an RF excitation source into a linac, for the purpose of accelerating electrons, the frequency of the RF source output must be adequately matched to the resonance frequency of the linac structure. Standing wave linear accelerators are somewhat more sensitive to the accuracy of the excitation frequency than are traveling wave accelerators, yet both are sensitive. In addition, a given linac's optimum excitation frequency is typically sensitive to temperature of the linear accelerator, as is well known. The output frequency of the RF excitation source can also change with environmental conditions such as temperature, and AFC (automatic frequency control) circuits are often used to maintain a good match between the RF source and the linac.

[0005] The maximum electron energy, and thus the resulting X-ray energy that can be obtained from a given linac structure is dependent upon the peak power coupled into it from the RF source, and also dependent upon the beam current within the accelerator at the time the RF power is applied, using known relationships. In addition, the Q of the linac can affect the performance of the system. For most linac-based systems, a high Q has been deemed desirable, typically between 5000 and 10,000.

[0006] In industrial applications such as cargo scanning, X-ray pulsing permits images of the

contents of a container to be created without opening the container. The high energy, MeV level X-rays from linac systems allow adequate penetration through large containers and their typical contents. The image of the contents of a container is typically a composite of a large number of scans, or image “slices” at different energies, in some cases 1000 to 10,000 or more such slices. As an example, a commercial cargo scanning system may pulse a linac at 100 to 400 X-ray pulses per second, as a truck passes through a scanner, and the images from each of those pulses are then composited to create the completed image. In some prior art cargo scanning operations, imaging systems that use linacs can scan between 24 and 150 trucks per hour depending upon the mode. [0007] While simple “gamma sources” using, for example, Cobalt 60, can be useful in cargo scanning, security scanning, and non-destructive test imaging, they incorporate radioactive materials that are controlled substances. Further, because the half-life of a material such as Cobalt 60 is just over 5 years, the Cobalt 60 must be replaced on a regular basis. The disposal must be done very carefully, ideally by trained and licensed personnel, as the material is extremely dangerous to humans, and is known to cause radiation sickness if handled improperly. Material such as Cobalt 60 is also considered by some to be a dangerous dirty bomb material. The upfront cost of a Cobalt 60 source may be less than a typical linac system, but the replacement and disposal associated with the use of Cobalt 60 becomes increasingly expensive and can pose serious dangers to humans. A linear accelerator system designed specifically to replace Cobalt 60 in cargo and security scanning applications would therefore be useful. Such a system would also be useful in non-destructive testing.

[0008] Different X-ray energies, such as 4 MeV and 6 MeV, are useful in cargo scanning because it permits the materials in the container to be differentiated. By comparing the images taken at 4 MeV with those taken at 6 MeV, steel can be distinguished from uranium, as just one example. Likewise, organic material, aluminum, lead, and so on, can be distinguished.

[0009] However, it is impractical to take a first scan of a container resting on the bed of a truck at a first X-ray energy, and then take a second, later scan of that same container at a second X-ray energy. Instead, it is more practical to interleave pulses of the different frequencies, so that a scan of both energy levels is taken in a single pass, for example by means of interleaved pulses of two energies in an ABABABAB pattern. This interleaving, however, can present challenges to a linac system since the performance of a linac with its resonant structure is highly dependent on a good frequency match. Some well known techniques for modifying the input parameters to a typical 6 MeV linac system, in order to obtain a 4 MeV output, involve either reducing RF power to reduce the maximum possible acceleration for electrons in the linac, or increasing the beam current to provide beam loading that ultimately reduces the maximum acceleration that each electron experiences, or a combination of these two effects.

[0010] These techniques have been sufficient when the time between energy changes is long, such as seconds or many seconds, but have been less effective when the time between energy changes is a fraction of a second. It is also possible to intentionally detune the resonance frequency of the RF source with respect to the linac—this has the effect of lowering the amount of RF power coupled into the linac, reducing the total possible acceleration for electrons. However, detuning can cause instabilities in performance, and is difficult to do in a fraction of a second if a mechanically-tuned magnetron is the RF source. A challenge with applying prior art techniques for changing linac energy on a rapid time scale in an ABABABAB fashion can be the undesired corresponding detuning of the RF source. For example, in addition to intentional tuning, reducing or increasing RF power from a magnetron typically results in a change in frequency of the RF output. If the frequency change is large enough, the linac's resonance is no longer matched to the frequency of the RF excitation pulse, and the system fails to operate. A pulse-to-pulse variation of RF amplitude or variation in RF frequency in a detuned condition will cause a greater change in linac output performance than operation at peak or tuned condition.

[0011] While the terms “4 MeV” and “6 MeV” are commonly used in the art, those terms typically

refer to the peak energies of the X-ray pulses, and the average energy can be less. These terms, and those of other energies, will be used herein with that same understanding.

[0012] Because of the sensitivity of linac-based systems to changes in frequency, the prior art has generally not been able to provide a fully optimized train of stable, interleaved pulses of different energies such as an ABABABAB (etc.) pattern, especially where the energy must be changed rapidly, from one pulse to the next, at a 2.5-millisecond basis or shorter such as required for a pulse rate of 400 or more pulses per second, and especially when that rapid change is done with a magnetron driven system

[0013] While magnetrons do not offer some of the advantages of klystrons, magnetrons such as the MG5193, MG6090, and MG7095 from supplier e2V have typically been used as RF sources for medical and cargo scanning applications. Similar magnetrons are available from National Japan Radio Corporation (NJRC). Unlike klystrons, magnetrons are not amplifiers, and the output frequency of a magnetron is adjusted by a mechanical tuner. This limits the ability to rapidly switch a magnetron system between higher and lower energy levels while still maintaining a frequency match with the linac, because the magnetron frequency will shift upon a power change and a mechanical tuner simply cannot be moved rapidly enough to support a 2.5 millisecond or shorter period between pulses. The duty cycle of such devices is typically low, for example approximately 0.1% for the MG5193, MG6090, and MG7095, such that pulse durations of a few microseconds, for example 2.5 microseconds, can be generated at up to 400 pps. For an MG5193, 4.5 turns of the mechanical tuner allow for about nine megahertz of tuning range. One drawback of a magnetron is that its mode and stability can become unfavorable if the magnetron is operated at a peak power too much different from its optimum or maximum peak power. The MG5193 can operate between about 1.5 MW peak and about 2.6 MW peak. The MG6090 and MG7095 can provide higher peak power outputs, such as 3 MW of peak power.

[0014] It is possible to change the power output of a magnetron rapidly through the use of a conventional high voltage capacitor-charging power supply together with a pulse forming network/modulator. With such a technique, interleaved high voltage ("HV") pulses can be supplied to the magnetron in an ABABABAB sequence. The result is that the RF power output of the magnetron can be varied rapidly as well, and also in an alternating fashion.

[0015] However, as noted before, changing the power output of the magnetron also causes a change to the output frequency of the RF pulse, such that the output frequency of the magnetron can be a mismatch for the linac. For example, the MG5193 and similar magnetrons can have a frequency shift of about 10 kHz per ampere, and they are typically driven at around 100 amps or more. Changing this current by many amperes may cause a significant detuning with respect to the resonance of a given linac. While AFC circuits can compensate for long term changes in frequency, such circuits are not intended to compensate for instantaneous pulse-to-pulse changes such as occur with a magnetron is driven with an alternating sequence that varies by several amps. As a result, such approaches can result in instability of the magnetron-linac system, and there has been a need for a system and technique for maintains stability while permitting the use of interleaved pulses of different RF powers to generate an interleaved pattern of X-ray pulses of different energies.

[0016] In addition to the challenges to prior art systems that result from variations in RF matching between the output frequency of a magnetron and the resonant frequency of the linac, the prior art has also had challenges in maintaining a consistent dose from pulse to pulse when interleaved pulses of different energies are generated. While dose control is known in medical systems, the need for ABABABAB sequences of different energies is not typically found in medical X-ray systems. Dose control in prior systems typically involves either changing the RF peak power coupled into the linac, or through beam loading, which involves changing the peak electron beam current into the linac, or both. For example, it is well known that increasing the peak beam current into an accelerator will reduce the energy of the electrons leaving the linac. The value of the beam current can be controlled by controlling the electron gun, but, if the maximum energy of the

electron beam is change, then the dose rate will also change absent taking steps to prevent that change. For example, using beam loading to change the X-ray energy from 6.5 MeV to 5 MeV can change the dose in some systems by a factor of two. However, for the use of X-rays to differentiate among materials, as required for cargo scanning applications, such changes in dose per pulse are undesirable. Instead, it is preferable that both energy and dose be controlled on a pulse-by-pulse basis even where the pulses are at different energies, such as energies that are different by more than 1 MeV.

[0017] While dose control in medical applications can be achieved through changes in repetition rate, this is not desirable for cargo scanning applications which depend upon having consistent repetition rates. As a result, there has been a need for a cargo scanning system which offers dose control on a pulse-by-pulse basis while offering stable sequences of interleaved pulses of different energies.

[0018] A further problem in cargo scanning applications is the intermittent nature of the operation of such scanning systems. The X-ray emission in a cargo scanning system is typically turned off after each scan is completed. For example, a first truck, carrying a container, can be scanned, and then the scanner is turned off. A next truck carrying a container arrives a few seconds later, or a minute, or some other indeterminate time period. As noted previously, the operating frequency of the linac changes with temperature. It is well known that the linac heats up during scanning, which requires that the output of the magnetron be adjusted to maintain a good match with the linac. While AFC circuits, which typically rely on feedback of the forward and reflected power of the linac, can maintain that good match when the linac system is being pulsed, there is no such feedback when the scanner is off and the linac starts to cool. As a result, typical prior art systems using conventional AFC may leave the tuner of the magnetron in the same position it was in when the scanner was turned off. Depending upon the frequency drift of the linac versus the RF source during the "off" period and the mismatch between RF source frequency and linac resonance, the system may simply not operate when scanning is restarted, or, more commonly, the energy output and/or dose output of the linac system will be less than intended. While this affects only the first few pulses, or 10 pulses or more after scanning is re-initiated, before the AFC circuit can achieve a good match, this lack of consistency in output energy and/or dose can affect the quality of the resulting images, and is therefore undesirable.

[0019] One prior art technique involves turning on the RF source to the linac in advance of enabling any beam current from the electron gun of the linac, to perform a partial warm-up of the linac beamline by the RF source prior to the electron beam triggering. In another approach, a typical AFC is used during the RF-only partial warm-up period, with an offset added to the tuning in advance of any expected electron beam triggering.

[0020] These warm-up approaches have the significant disadvantage of potentially generating some amount of X-rays even when the electron gun is not being pulsed. Such X-rays can be generated by virtue of the high electric fields in the RF-pulsed linac pulling electrons from the electron gun and cavity walls. Another disadvantage is the consumption of average power that is not used for the generation of X-rays. These approaches are inefficient and therefore undesirable.

[0021] As a result, there has been a need for a linac-based X-ray scanning system that provides consistent pulse energies even for the initial pulses of a sequence despite intermittent operation and "off" periods of indeterminate duration.

SUMMARY OF THE INVENTION

[0022] The present invention overcomes the aforementioned limitations of the prior art by providing a stable sequence of interleaved pulses of different energies, while at the same time providing consistent and precise pulse-to-pulse dose control.

[0023] In one aspect of the present invention, a series of X-ray pulses of at least two different desired energy levels is created by providing an RF power generator, typically a magnetron, which supplies to a particle accelerator such as a linac RF pulses of at least two different controllable

powers at two different corresponding rf frequencies. The Q of the linac is designed to be sufficiently low that the different RF frequencies of the pulses from the RF power generator remain within the resonance bandwidth of the linac. At least one gun driver supplies at least two different controllable current pulses to the electron gun of the linac, with the pulses of each different current synchronized to occur within an envelope created by an RF pulse of a corresponding power. The resultant electron pulse from the electron gun of the linac is accelerated until it strikes a target, for example tungsten, to generate an X-ray pulse at one of the desired energy levels. The dose of each pulse is adjusted by altering the duration of the current pulse of from the electron gun, which permits the X-ray pulses of different energy levels to supply the same dose per pulse, or different doses adjusted to a desired ratio.

[0024] In a related aspect, in some instances it is desirable to scan a first portion of an object or a vehicle using pulses of a first dose, and then scan a second portion of the object or vehicle using pulses of a second dose. For example, when scanning a truck, the driver is frequently in the cab of the vehicle during the scan. It is desirable in at least some instances to scan the cab where the driver is located with pulses sufficient to provide good imaging, without exposing the driver to the same dose as is desirable for scanning a cargo-containing area such as a trailer, cargo container, or rail car. In an embodiment of the present invention, a pulse train of lower dose can be generated while scanning the cab, and a pulse train of higher dose can be generated while scanning the trailer. The pulse train of lower dose can be alternating energy levels, such as ABABABAB, or can be a sequence of pulses of a single energy level, such as AAAAAA or BBBBBB, or can be a sequence of pulses having a different energy, such as CCCCCC, or can be any combination of these. In a still further related aspect, it will be appreciated that pulse trains of alternating energies can comprise more than two energy levels as, for example, ABCDABCDABCDABCD, or any combination of energy levels made desirable by the specific implementation as long as the various pulses are consistent with the bandwidth of the linac.

[0025] In some embodiments comprising a single energy pulse but with higher and lower dose modes, a single grid gun driver can be implemented to simplify design. Unlike typical prior art linac-based systems, in at least some embodiments, the dose rate used for scanning the cab or similar region of a vehicle is much lower than the dose rate used for scanning the cargo-containing area, to maintain safe levels of radiation for a driver or passenger. Typical prior art linac-based systems do not have the capability of varying dose by the necessary magnitude and attempt to scan regions of interest at very different X-ray dose levels by using a non-linac X-ray source for a lower dose region and using a linac-based X-ray source for higher dose cargo scanning.

[0026] While linac energy levels of 4 Mev and 6 Mev are desirable for many applications, including but not limited to cargo scanning, another useful range of linac energies for cargo scanning is between approximately 2 Mev and 4 Mev. These energy levels are sufficient to equal or exceed the performance of systems using Cobalt 60. In addition, the lower energy levels permit the exclusion zone to be smaller, and, in some instances, passengers can be permitted to stay within a vehicle that is being scanned. Lower energy systems can also be lighter with less shielding, and in some embodiments can be mounted on a mobile platform.

[0027] In a different aspect of the invention, compensation is provided for the changes in performance in both the linac and the RF power generator as the result of thermal and other influences caused by intermittent operation. For example, in a cargo scanning application, the linac system is operated to generate X-rays during the period that a cargo container passes through the scanner. The scanning function is then turned off, for example by disabling power to the RF power generator, until the next container is moved into position for scanning. The scanning function is then restarted. The "off" period is of uncertain duration because it depends upon when the next container is properly positioned. As a result, the performance parameters of both the RF power generator and the linac can vary significantly, to the point that a mismatch between the frequency of the RF generator pulses and the linac resonance can occur which may prevent stable system

operation altogether.

[0028] To prevent such mismatches resulting from the intermittent operation of the linac system, the present invention provides a system and method for adjusting the frequency of the RF power generator during periods when the scanning function is disabled, such that, when scanning is re-enabled, the frequency of the RF power generator has been adjusted to maintain a substantial match with the changed resonance of the linac resulting at least in part from the cooling of the linac while scanning was disabled. In addition, in some embodiments, ambient temperature is sensed and incorporated into a cool down compensation system and method. In at least some embodiments, the compensation results in the dose of the first pulse after re-enabling being within ten percent of the average dose during the period when scanning is enabled, and, depending upon the embodiment, can be within two percent or one percent of the average dose.

[0029] Depending upon the embodiment, the RF power generator can be either a magnetron or a klystron. In those embodiments using a magnetron, the frequency of the magnetron is adjusted by mechanically adjusting the tuner with a stepper motor, with the AFC causing motion of a number of steps from the “home” setting, where the home setting is the optimized setting of the tuner to match the RF output frequency with the resonance of the linac when the linac is operated at a sufficiently low repetition rate that thermal influences do not greatly change its resonance. After a period of operation, the AFC will move the tuner to an optimum position for linac performance. During an “off” period of indeterminate duration, the cool down compensation system and method moves the tuner in the absence of feedback in order to maintain a match between RF source frequency upon restarting pulsing, and that of the linac. In a preferred embodiment, the number of steps of tuner correction may correlate to the duration of the “off” period, and, if the off duration is long enough, the correction eventually causes the tuner to return to the home position. For embodiments using a klystron, the RF output frequency is adjusted electronically.

Description

THE FIGURES

[0030] The foregoing summary of the invention, as well as additional aspects and features, will be better understood from the following detailed description, taken in conjunction with the appended Figures, in which:

[0031] FIG. 1 is a block diagram of a linac-based X-ray system in accordance with an embodiment of the invention.

[0032] FIG. 2 illustrates in cross-section a linac in accordance with an embodiment of the invention.

[0033] FIG. 3 illustrates the relationship between RF pulses and electron beam pulses in accordance with an embodiment of the invention, including the use of beam pulse duration to achieve pulse-to-pulse dose control.

[0034] FIGS. 4A-4E are timing diagrams showing an interleaved sequence of stable pulses in accordance with an embodiment of the invention, including dose control.

[0035] FIG. 5 is a block diagram of a dual mode gun driver in accordance with the invention.

[0036] FIGS. 6A-6D are timing diagrams illustrating the cool down cycle which results from intermittent scanning operation, and the improvement resulting from the cool down compensation system and method of the present invention.

[0037] FIG. 7 illustrates in block diagram form an alternative embodiment of the invention which offers dose-to-dose pulse control.

[0038] FIG. 8A shows in timing diagram form the pulse shapes of FIG. 3, and distinguishing higher and lower energy pulses with a “normal” higher dose.

[0039] FIG. 8B shows in timing diagram form the pulse shapes corresponding to those of FIG. 8A,

but in reduced dose mode for both higher energy pulses and lower energy pulses.

[0040] FIG. **9A** shows the reduced dose pulse that results from the waveforms of FIG. **8B**, in a pulse train of the form ABABAB.

[0041] FIG. **9B** shows the reduced dose pulse that results from the waveforms of FIG. **8B**, but where the reduced dose pulses C are a single energy different from either A or B.

[0042] FIG. **10A** shows a combined pulse train of a single energy level for the reduced dose portion and a single energy level for the higher dose portion.

[0043] FIG. **10B** shows the transition time between the reduced dose mode and the normal dose mode for the single energy system of FIG. **10A**.

[0044] FIG. **11** shows a single grid gun driver in accordance with an aspect of the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0045] Referring first to FIG. **1**, an X-ray scanning system according to an embodiment of the invention is shown in block diagram form at **100**. In such an embodiment, external power and signals **105** are received by a control processor **110**. Included among the external signals are, typically, one or more trigger signals indicating that the user desires to scan an object, for example a cargo container on a vehicle passing before the scanning system. The control processor **110** controls, directly or indirectly, the operation of the remaining functional blocks shown in FIG. **1** by virtue of signals sent on internal bus **115**, which, for simplicity, is shown combined with internal power.

[0046] In response to the trigger signal(s), the control processor **110** sends, depending upon the implementation, a plurality of signals to initiate generation of an X-ray pulse. In particular, the processor **105** sends control signals to a high voltage power supply **120** and an associated modulator **125** which receives the output from the supply **120**. The supply **120** can be, for example, a Lambda LC1202. The output of the modulator **125** supplies a high voltage output to a pulse transformer **130**, typically immersed in an insulating tank for purposes of electrical isolation. An aspect of the modulator is that can vary the voltages from one pulse to the next, and can operate at pulse durations of 2.5 usec or less, to permit operation at 400 pulses per second. The modulator may incorporate a pulse-forming network or PFN. A heater power supply **135** is associated with the tank and supplies the magnetron **140** or other suitable RF power source. The pulse transformer **130** supplies high energy pulses, for example 30-50 kV at 100-110 amps, to a magnetron **140** or other suitable RF power source. One suitable magnetron is the e2V model MG5193, which has an output of 2.6 MW at the normal S band frequency of 2.998 GHZ. Another is the MG7095, also from e2V. Still other similar magnetrons are available from NJRC. The specific magnetron frequency is controlled by a mechanical tuner **145**.

[0047] The magnetron **140** outputs an RF power pulse, indicated at **150**, at the frequency determined by the tuner **145**. As explained in greater detail below, the pulses received by the magnetron can be of different, pre-selected voltage and currents, thus causing the magnetron to output pulses of different, pre-selected RF powers, for example, pulses of 40 kV and 45 kV at 100 amps and 110 amps, respectively. Because of the nature of the magnetron, the different powers of the RF pulses also affect the frequency of the output pulse, again as explained in greater detail below. The RF power pulses pass through an arc detector **155**, an isolator **160**, and then to a linear particle accelerator (sometimes "linac" hereinafter) **165**. Suitable isolators are available from Ferrite Incorporated. Conventional S-band waveguide **157** is used between magnetron and linac. Depending upon the requirements of the particular linac, for example of the type shown in FIG. **2**, the pulses received from the transformer **130** can be in the range of, for example, 35-50 kV. The linac **165**, which typically has an effective Q in the range of 2000-4000, but in any event less than 5000, receives the RF pulse. The tuner **145** is adjusted so that the RF pulses from magnetron **140** are within the resonance bandwidth of the linac **150**. The pulses from the magnetron are, in an embodiment, substantially in the range of 2.5 MW, or between 2.0 MW and 3.0 MW.

[0048] In the embodiment of FIG. **1**, the control processor **110** sends a control pulse to the

modulator **125**, and it sends a synchronized control pulse to a dual mode electron gun driver block **170**. The timing of these control pulses may be individually optimized. The dual mode electron gun driver block **170**, explained in greater detail hereinafter in connection with FIG. 5, drives an electron gun **175**, the cathode of which is within the vacuum envelope of the linac **165**. The gun **175** can be a triode gun design. In an embodiment, the pulses of beam current from electron gun **175** launch electrons into the cavities of the linac. The cathode voltage is substantially in the range of -10 to -20 kV. In an embodiment it is -20 kV. By timing the electron pulses at the linac to optimally synchronize with the electric field created by the RF pulses from the magnetron, as discussed in greater detail hereinafter, the electrons are accelerated by the linac to a desired energy level, typically in the range of two to ten or more MeV with, for at least some embodiments, a separation between the energy levels of approximately 1 MeV or more between sequential pulses in a rapidly pulsed ABABABAB pattern. The pulses are directed toward a target **180**, for example tungsten, which, when hit with the pulse of accelerated electrons, emits pulses of X-rays. As discussed below in connection with FIGS. 3 and 4A-4E, the RF pulses are, in an embodiment, somewhat longer in duration than the electron gun pulses, such that the RF pulses can be thought of as creating an envelope within which the beam current pulses occur. To control dose, the duration of the beam current is selected by the control system **105**, or can be pre-set during manufacture.

[0049] Because the transmission characteristics of the linac and magnetron vary with temperature and other environmental factors, an AFC circuit **185** detects forward and reflected power from the linac, using dual directional couplers **190**, and in turn controls the tuner **145** to ensure a continuing match between the linac and the magnetron. in a manner known to those skilled in the art.

[0050] In another aspect of the invention, in some embodiments, the AFC circuit **185** also includes cool-down compensation, discussed in greater detail hereinafter in connection with FIG. 6. Because X-ray scanning systems operate intermittently, and the duration of the “off” periods is essentially unpredictable, the cool down compensation of the present invention operates the tuner during periods when the scanner is off to maintain a good match between the linac's resonance bandwidth and the output frequency of the magnetron, thus providing improved operational stability.

[0051] In addition, ancillaries **195** connect to the arc detector **155** and an ion pump **200** that feeds the linac **165**, both in a manner known to those skilled in the art. Finally, a cooling system **205** cools portions of the system in a manner known in the art, for example, the modulator, the pulse transformer tank, the linac, the target, and the isolator, as indicated by the dashed line **210**. Suitable cooling systems are available from OptiTemp, and can be chosen dependent upon temperature and cooling requirements of the linac system.

[0052] With reference to FIG. 2, a linac **165** in accordance with an embodiment of the invention can be better understood. The linac **165** is configured to provide X-ray pulses of different and controllable energies and doses in an alternating fashion. The linac comprises an electron gun **175**, which can comprise a gridded electron gun. The beamline **220** is comprised of the main cylindrical section, and is comprised of cavities **225** that have been designed in order to accelerate electrons from the cathode of the electron gun up to speeds that correspond to energies in the MeV range. The cavities of the beamline can be inline coupled or side-coupled, and the cavities are also designed for resonance with the RF source, such as the magnetron **140** in FIG. 1. A water-cooled target assembly **230** is attached to the linac beamline, and houses a target **180**, which is made substantially of tungsten, or other suitable material. In an embodiment, the linac is designed to accelerate electrons up to 6 MeV in energy and can also accelerate electrons to 4 MeV in energy on an alternating pulse, interleaved basis, for the purpose of generating X-rays with similar maximum energies per pulse. An ion pump **200** is connected to the vacuum envelope of the linac in a conventional manner. An RF window **235** is attached to the linac beamline, for the purpose of input coupling of RF power. Between the RF window **235** and the beamline **220** is a coupling iris **240** or opening, the size of which can be adjusted in order to assist in tuning the effective Q of the linac assembly.

[0053] As discussed above, one aspect of at least some embodiments of the present invention is to generate interleaved sequences of X-ray pulses of at least two differing energies, for example 6 MeV and 4 MeV. In an embodiment, and with reference to FIGS. 3 and 4, the interleaved sequence of pulses rapidly alternates between the higher energy and the lower energy pulses, e.g., ABABABABAB, although other embodiments can provide different sequences of interleaved pulses, such as AABB or any other desired sequence. For purposes of clarity only, the alternating, or ABABABAB, sequence will be described in greater detail hereinafter, as those skilled in the art will understand how to provide other sequences, given the teachings herein. It will also be appreciated that more than two energy levels of X-ray pulses can be generated with the current invention as long as the frequency shifts of the magnetron pulses remain within resonance bandwidth of the linac and are properly matched to the beam current. For purposes of illustration, use of the present invention to generate an AB sequence is explained below, with reference to FIG. 1 and the timing diagrams of FIG. 3.

[0054] To generate an alternating sequence of lower and higher energy X-ray pulses, the control system 110 instructs the modulator 125 to supply a different level of power, to the pulse transformer 120 for each different energy level of pulse. This, in turn, varies the power supplied to the magnetron 140, such that the alternating higher and lower RF pulses are supplied by the magnetron to the linac, as shown in FIG. 3 at 300 and 305. By varying the power to the magnetron, the frequency of the output of the magnetron changes, for example by approximately 10 kHz per amp, or 100 kHz when the current supplied to the magnetron varies from 110 amps for 6 MeV to 100 amps for 4 MeV, as an example. As discussed above, the Q of the linac is configured to provide a resonance bandwidth of sufficiently more than 100 KHz, so that both the higher and lower frequencies output by the magnetron remain within the operating range. Again as discussed above, in an embodiment, the Q of the linac is preferably in the range of 2000-4000, and a Q of 3000 has been demonstrated to operate well. In any event, the Q is less than 5000.

[0055] As shown in FIG. 3, within the pulse duration of the magnetron pulses 300 and 305, the electron gun driver also generates current pulses of different energies, as shown at 310 and 315. The A pulse is of a desired electron and X-ray energy and X-ray dose, and the B pulse is of a different electron and X-ray energy and X-ray dose that is at least 1 or 1.5 MeV different from the energy associated with the A pulse. The magnetron current pulses are delivered by the pulse forming network and modulator of a preferred embodiment of the invention, and have a duration that is set by the design of the pulse forming network within the modulator 125 shown in FIG. 1. Other modulators could be used where the magnetron current pulse duration can be changed. In an embodiment, the duration of the current pulse is between 2 and 2.5 microseconds for both A and B pulses, and the repetition rate of the pulses is on the order of 100 to 400 pulses per second, or whatever is allowed by the maximum average power dissipation and delivery that a given magnetron can tolerate. For a faster magnetron, a rate of 1,000 pps, and, at the other end of the range, a rate of 1 pps can also have utility in some applications.

[0056] The higher current magnetron pulse is typically correlated to the higher energy X-ray pulse, and the lower magnetron current pulse is typically correlated to the lower energy X-ray pulse. The high and low currents cause a higher and lower RF peak power to be delivered to the linear accelerator. The difference between these currents is limited in order to limit the magnetron frequency shift that occurs as a result of different voltage and current applied to the magnetron, in order that both the A and B magnetron frequencies are both well matched to the resonance bandwidth of the linac while still obtaining at least 1 MeV of difference between the X-ray energy of the A pulse and that of the B pulse, when pulsing in an interleaved sequence.

[0057] As discussed in connection with FIG. 5, below, the gun driver can be controlled from the control system to alter amplitude, duration and timing of the beam current pulses. The gun current magnitudes are controlled by the gun driver, so that different beam currents for the A pulse and the B pulse are controlled. The combination of different RF power and different beam currents allows

control of the energy of the A and the B pulses, in accordance with the method of the present invention. In an embodiment, the energy difference between the A and B pulses is 2 MeV, although in other embodiments the difference can range from 1 to 5 MeV. Further, the gun current pulse durations of the A pulses and the B pulses are independently controlled by the gun driver in order to control the X-ray dose on a pulse-to-pulse basis. The duration of the voltage pulse to the grid substantially controls the duration of the gun current pulse, and therefore the duration of the X-ray pulse. If all other parameters are fixed, and only the pulse duration parameter is adjusted, then the dose per pulse will be adjusted while the X-ray energy for a given pulse will remain the same. In an embodiment, the dose of the A pulse and the dose of the B pulse are substantially equal. In another embodiment, the ratio of the A dose and the B dose is adjusted to a desired ratio other than one. [0058] In addition to the adjustment of gun current pulse duration to control the dose per pulse, the timing of the start of the gun current pulse with respect to the RF pulse can also be adjusted. In a preferred embodiment, the leading edge of the gun current pulse begins early in the “flat” part of the RF waveform. In other embodiments, the leading edge of the gun current pulse can be started later with respect to the “flat” part of the RF waveform.

[0059] The overall AB sequence can be appreciated from FIGS. 4A-4E, in which FIG. 4A illustrates two sequences of X-ray pulses with an “off” period of indeterminate duration in between. FIG. 4B illustrates the dose per X-ray pulse for the higher energy X-ray output, while FIG. 4C illustrates dose per pulse for the lower energy X-ray output. Both FIGS. 4B and 4C illustrate a range of dosage amounts with a variance which represents acceptable stability, typically about 10 percent or less, or 2% or less, or 1% or less. FIG. 4D illustrates the current supplied to the magnetron for each of the lower and higher energy pulses, while FIG. 4E illustrates the beam current supplied to the linac for the higher and lower energy pulses. It will be appreciated that the beam current is lower for the higher energy pulses than for the lower energy pulses in an embodiment. In one embodiment of the invention, the peak beam current is controlled to between approximately 25 mA and 125 mA when high energy pulses are desired, and the peak beam current is controlled to between approximately 125 mA and 250 mA as associated with the low energy pulses. Other beam currents are selectable with the linac system of the invention. In an embodiment, the beam currents can be precisely tuned in conjunction with the RF power in order to precisely provide a desired X-ray energy, and thus the X-ray energy can be alternated on a rapid pulse to pulse basis in an ABABABAB interleaved manner.

[0060] As discussed above, the pulses generated by the electron gun driver are adjustable in amplitude, duration and timing, which permits the beam current pulses to be synchronized and matched with the RF pulses to generate X-ray pulses of different, controllable energy levels, with consistent, controllable dose per pulse. FIG. 5 illustrates an embodiment of a dual mode gun driver suitable for generating pulses of 2.5 microseconds or less, suitable for cargo scanning functions which typically operate from 100 to 400 pulses per second, in accordance with an embodiment of the invention. Higher repetition rates are possible if standard electrical and thermal parameters are considered, as will be appreciated by those of skill in the art.

[0061] The gun driver 500 shown in FIG. 5 essentially comprises two independently controllable driver modules, one for the high energy pulses, indicated at 505, and another for low energy pulses shown at 510, where each module connects to the grid of the gun 175 (FIG. 1) at the appropriate times through control of high voltage switching transistors, and supplies the appropriate pulse at appropriate but different times as shown in FIG. 4E. The high energy module 505 comprises a heater supply 515 and high energy grid top power supply 520, with the output of the grid top supply 520 connected to the grid output through a switching transistor indicated at 525. The low energy module 510 comprises a low energy grid top power supply 530 and a grid bias supply 535, and the low energy grid top power supply connects to the grid via a switching transistor 540 and a diode 545, while the grid bias supply connects to the grid through a switching transistor 550. The gun driver also comprises a high voltage filter module 555 and a high voltage power supply 560 in

a conventional manner.

[0062] The performance of the gun driver modules, including the high voltage switching transistors, is controlled from the control system **110** and bus **115**, as more generally shown in FIG. **1**. Feedback signals from each module are also used, as is common in the art of electronics. The grid modulating modules **505** and **510** are referenced to the cathode voltage, as shown, and apply voltage to the grid of a triode gun that is part of the linac, shown at **175** in FIG. **1**. As is known in the art, the control of grid voltage is a common method for controlling the gun current emitted from the cathode of a triode or gridded electron gun. In the preferred embodiment, one grid modulator is used to control the grid voltage associated with the A pulse, and the other grid modulator is used to control the grid voltage associated with the B pulse. In another embodiment, a single grid modulator can be used if the grid voltage can be changed accurately to desired values on a pulse-to-pulse time basis. The grid voltages are controlled in order to control the gun current that is launched into the linac, as the magnitude of the beam current in the linac is a function of the gun current. The independent control of the amplitude of the beam current, combined with the independent control of the RF power, is used to control the X-ray energy on a pulse to pulse basis. [0063] The duration of the grid pulse is used to control the dose of a given X-ray pulse. As discussed above, a shorter pulse yields a lower dose per pulse, whereas a longer pulse corresponds to a larger dose per pulse.

[0064] Referring next to FIGS. **6A-6D**, a different aspect of the present invention can be appreciated, by which pulses of consistent energy are provided even with a “cold start” or after the linac system in the scanner has been off for an indefinite period. More specifically, FIGS. **6A-6D** depict both the effect of classic AFC techniques as well as the novel cool down compensation technique of the present invention that provides stability from the first pulse in an intermittent pulsing application. Since the invention outlined above uses standard prior art AFC techniques, the interleaved ABABABABAB etc. output can be stable over long time periods, for example, seconds, and minutes. Longer durations are possible as well. Prior art AFC techniques require at least one feedback signal proportional to the reflected RF power from the linac to insure proper mechanical tuning of the magnetron, as in the case of a preferred embodiment, and as is well known. In the case of a klystron, tuning is accomplished via an input RF driver, but the AFC still requires feedback representative of the output of the RF source with respect to the linac resonance. Since feedback techniques are used, many RF pulses (and therefore X-ray pulses in most systems) are required before the AFC subsystem has sampled enough information and adjusted the magnetron mechanical sufficiently that optimal and stable X-ray pulses are generated. A conventional AFC system alone is not sufficient to allow highly stable operation from the first X-ray pulse after an “off” period without RF feedback. In an embodiment of the invention, in order to take the same linac system with AFC and provide optimal and stable pulsing from $t=0$, or what is sometimes called a cold start, a cool down compensation (“CC”) algorithm has been invented and is used.

[0065] With classic AFC circuits, the AFC compensates for thermal effects or long term drift effects that cause a drift in the X-ray pulse performance. If a drift or deviation in performance takes place over many pulses, and if it is due to a correctable frequency mismatch between the RF source and the linac resonance, then the classic feedback techniques used in prior art AFC subsystems can be used to stabilize the system. However, AFC designs and methods typically require many pulses to correct the position of the magnetron tuner and so, if only the AFC is used and if the pulsing is intermittent, that is, shut off for an irregular period of time and then turned back on, the dose and/or energy per X-ray pulse for some number of initial pulses can deviate significantly from the intended dose or energy per pulse.

[0066] FIG. **6A** depicts the operation of the magnetron in the context of a classic heating and cooling curve as shown in FIG. **6B**. The shape of that curve is generally associated with the deposition of heat during pulsing, and then the dissipation of heat once pulsing stops. This is a

classic phenomenon known to those of skill in the art, and when heating is occurring due to pulsing, a classic AFC can help maintain frequency match between a magnetron and a linac. However, when pulsing is turned off, then the system will cool, but the AFC cannot adjust the tuner in the same manner because it receives no feedback signals. The conventional feedback signals like forward RF power and reflected RF power do not exist for the AFC during “off” or non-pulsing periods. When a feedback signal is removed, prior art AFC subsystems have no input information with which to drive a motor or tuner, and may maintain the motor and tuner position that was appropriate for the most recent pulse, or some other position, but that is not appropriate for the next pulse at some indeterminate future point in time. This can result in non-optimum X-ray output or output at unintended values once the system is restarted, because the characteristics of the linac have changed during the off period.

[0067] In an embodiment of the invention, cool down compensation logic is provided in the AFC circuit (**185** in FIG. **1**) that causes the position of the tuner to be changed during non-pulsing periods without the need for feedback, so that the output frequency of the magnetron remains properly matched to the linac resonance characteristics despite the cooling that occurs during an “off” period. In an embodiment, this technique can achieve stabilities significantly better than 10%, and in some instances approximately 1%. The cool down compensation logic can be a look up table (LUT), an algorithm, or other implementation, and is addressed by the AFC circuit when no pulses are detected for a predetermined period of time, for example, one second.

[0068] FIG. **6C** is a timing diagram that shows the movement of the tuner during cooling periods shows the changing tuner position that follows, and compensates for, the cooling that occurs when scanning is not active. FIG. **6D** is a plot of dose per pulse, and illustrates that, as the result of the cool down compensation, the first pulses after a restart are substantially identical in energy and dose as the pulses that occurred during prolonged scanning with the AFC circuit active. In an embodiment, the AFC circuit with cool down compensation logic drives a stepper motor, and the stepper motor moves the tuner on the magnetron in a conventional manner.

[0069] The cool down compensation data used to populate a LUT or define an algorithm, as discussed above, can be developed as follows. For a given RF value, a proper magnetron starting or “home” tuner position is determined by very low repetition rate operation of the linac system, at the desired energy and dose per pulse. For example, this home tuner position can be determined at 1 to 10 Hz operation, and can be set by an operator who optimizes system performance vs tuner position, either manually or with a classic AFC. The very low repetition rate simulates a scenario where very little heating is occurring. This home tuner position is recorded in system memory. This type of operation simulates the behavior of a linac system with very little stored heat in either the linac or magnetron. In a preferred embodiment, the magnetron tuner is driven by a stepper motor, and the number of steps away from a motor reference point or mechanical stop is used as a proxy for tuner position. As an example, the home position may be determined to correspond to 50 steps away from the reference point or mechanical stop. A tuner position associated with full power and full repetition rate operation may achieved by 100 or 150 steps away from the reference point or mechanical stop, as an example

[0070] For the CC technique of the present invention to cause optimal tuning during non-pulsing, a calculation or look-up table is generated with motor positions that correspond to substantial matching between magnetron frequency and linac frequency during the time intervals where no pulsing is commanded. After very long off periods, like minutes, the proper motor position corresponds very nearly to the “home” position. After long “on” periods, for example after minutes of operation, the motor position determined by a classic AFC will be many steps away from the home position, in order to tune the magnetron to the linac resonance in a warmed state. For “off” periods of varying durations, the optimal motor position will be somewhere between these two positions, and must be determined.

[0071] A variety of techniques can be used to determine optimal motor and tuner position in the

absence of pulsing. For an embodiment of the system, an outline of one procedure that can be used to develop the tuner positions during cool down, and thus the entries for a LUT or other correction technique, is as follows: [0072] 1. The system is operated at full energy and power, so for example, the magnetron output is 2.6 MW, and it is pulsed at 400 pps with pulse durations of about 2 microseconds for the flat-top of the RF pulse. This mode corresponds to the maximum heat deposition to the system in an embodiment. A standard AFC circuit maintains the match between the magnetron frequency and the linac resonance during this pulsing period, with a time constant on the order of 0.5 seconds. Near steady state can be achieved in times on the order of a few minutes for this approach. [0073] 2. The system is turned down to 10 pps abruptly, and the AFC is allowed to tune the magnetron tuner in a standard way, typically by driving a stepper motor. Every 5 seconds, an operator or a data acquisition system records the position of the stepper motor, over a period of several minutes. [0074] 3. A plot of stepper motor position is created, which shows steps on the Y axis, and time on the horizontal axis. [0075] 4. The plot in step 3 above can be fit to an exponential decay with a time constant, or it can be used to create a look-up table. The calculated exponential decay can be used to calculate the proper tuner position as a function of its most recent position; in a preferred embodiment, the farther a tuner has been stepped from its zero heat or “home” position, the larger its correction per unit time towards the zero heat or “home” position should be.

[0076] In an embodiment, a rate of change in terms of steps per second towards home may be calculated or measured from the data collected in steps 1 to 3 above. The rate of change can be plotted as a function of the steps away from home. In a preferred embodiment, the optimal rate of change is largest for position changes that are large with respect to the home position. With the CC active, for any position of the motor that drives the magnetron tuner, there is an associated rate-of-change in steps/second corresponding to that position. This rate of change can either be calculated by determining how many steps the AFC moved between intervals in the procedure 1 to 3, or it can be derived from a fit to the exponential curve to the position vs time data collected in steps 1 to 3. Alternatively, a look-up table can be created.

[0077] In an embodiment, an optimal expression of the steps per second as a function of steps from home can take the form

$$Y=a*e^{sup.(b*x)}$$

where Y is the optimal steps per second, x is the steps from home, and a and b are constants determined by a mathematical fit to experimental data for the linac system in an embodiment. In general, a is a coefficient representing the magnitude of the exponential equation, and b is a coefficient for curve-fitting. For example, in an embodiment where the stepper motor has on the order of 100 steps of range, a can be 0.13, while b can be 0.034. Those skilled in the art will recognize that these coefficients can be different numbers depending on the thermal behavior of the system, or the behavior of the system in response to other parameters. When the CC is used to move the stepper motor that moves the magnetron tuner, and the CC uses the formula, the pulsing performance in an intermittent pulsing scenario was very nearly optimal for every pulse, as shown in FIGS. 6D and 6E.

[0078] In an alternative embodiment, the following procedure can be used: [0079] 1. Warm up at 400 pps for 3 minutes, note warmed-up tuner motor step position with respect to home (position as controlled by classic AFC). [0080] 2. Shut off for 5 seconds, turn back on, record tuner motor step position after 2 seconds (position as controlled by classic AFC). [0081] 3. Warm up again at 400 pps to same warmed-up tuner motor step position as #1. [0082] 4. Shut off for 10 seconds, turn back on, record tuner motor step position after 2 seconds (position as controlled by classic AFC). [0083] 5. Warm up again at 400 pps to same warmed-up tuner motor step position as #1. [0084] 6. Shut off for 15 seconds, turn back on, record tuner motor step position after 2 seconds (position as controlled by classic AFC).

[0085] Continue this sequence for up to 2 minutes of shut off time. Plot the data as tuner motor position in “steps-from-home” (y axis) vs “cool-down time” (x axis). [0086] 7. Fit an exponential to that data. [0087] 8. Take the derivative of that data, and plot it on the same graph, which will now depict both the exponential fits to “steps-from-home” and “steps/second” vs “cool-down time”. [0088] 9. Create a new plot of “steps/second” vs the “steps-from-home”.

[0089] In an embodiment, this table created from such techniques is used by the AFC with CC during customer-triggered intermittent operation to provide the rate at which the tuner motor should be moved any time the RF is turned off, based upon the last position where the tuner motor was left by the AFC. In this embodiment, the AFC controls the tuner motor position while RF and X-rays are pulsing, and the CC controls the tuner motor position when RF and X-rays are turned off.

[0090] In some cases, further improvement resulting in better pulse-to-pulse stability is desirable. In the application of cargo scanning this can facilitate better imaging, and in the application of radiation oncology this can facilitate more stable dose delivery. For example, an RMS stability of the X-ray dose per pulse of nearly 1%, or less than 1%, is desirable. In the case of the linac operating in dual energy on an alternating basis, the dose per pulse at one energy may be less stable than the dose per pulse at a different energy. To enhance stability of the dose beyond what is achieved by AFC alone, additional techniques can be used. One technique, which uses substantially the hardware structure shown in FIG. 1, takes a digital output from the AFC circuit **185** during X-ray pulsing bouts in order to generate a correction signal for the programming voltage for the modulator **125** that drives the magnetron **140**. The programmed voltage for steady state magnetron operation is typically a fixed value, for example, a high voltage power supply programmed voltage of 20 kV will result in a PFN output of approximately 10 kV which results in a pulse to the magnetron of 45 kV (after a 4.5:1 pulse transformer.) This results in an RF output of approximately 3 MW, which in combination with a specific gun current will result in a specific desired energy, such as 6 MeV. Using the AFC circuit's digital output of the AFC stepper motor position, a correction value to the normally fixed programmed voltage can be created, using the reported AFC stepper motor position with respect to “home” or zero position. In a preferred embodiment, the correction value can be positive or negative, and is added to the steady state programmed voltage that corresponds to one of the desired pulse energies. In a preferred embodiment, this correction value contains a term proportional to the AFC stepper motor position with respect to the “home” position that corresponds to tuner position with no system heating, and this correction value also contains a fixed offset term. Either term of the correction value can be positive or negative, depending on the direction of desired correction to the pulse-to-pulse X-ray dose over a typical 30 to 60 second scan. The correction value can be added to the programmed voltage to just one of the alternating energy pulses (either the high or the low energy), or to both, or a differently scaled correction value can be generated for each of the energy settings. By adjustment of the correction value, enhanced X-ray stability over the duration of a typical scan is achieved, resulting in an improved image in a security scanning system.

[0091] Another method to stabilize the X-ray dose per pulse is by measuring target current on a pulse-to-pulse basis. One approach for accomplishing this is where the target assembly has been electrically isolated from the linac body, and current due to the electron beam striking the target is measured with a simple circuit. Peak target currents of between 25 mA and 250 mA during a pulse are typical values, which allows generation of a measureable signal voltage across a resistor in series with the target current. If the electron energy is constant, the dose for a given X-ray pulse or series of pulses is directly proportional to target current. Therefore, if electron energy is constant, a feedback loop can be implemented using target current as the measured parameter that is a proxy for X-ray dose. The feedback loop stabilizes target current (and thus X-ray dose), on a near pulse-to-pulse basis by adjusting the high voltage power supply program voltage (the voltage that drives the magnetron, as discussed above). Another method to achieve this effect is to measure the

integrated target current over an individual pulse, in order to calculate the total charge delivered in an individual pulse. In this alternate method, using integrated target current per pulse, a servo loop that adjusts pulse duration at the target is used to stabilize the integrated target current per pulse with respect to a desired value. In a system with a triode or gridded gun and an independent triode gun driver, a straightforward way to control the duration of a pulse at the target is by controlling the duration of a pulse to the grid as delivered from the gun driver. Certain solid state modulators may also facilitate this form of stabilization. However, this approach can suffer from certain instabilities if a change in energy occurs.

[0092] A still further alternative, shown in FIG. 7, offers some advantages because it measures actual dose on a pulse-to-pulse basis rather than the integration-based averaging performed by many other designs, and, by means of a feedback loop, uses the measured dose to stabilize the successive generated doses. The feedback loop can, for example, control either the voltage level at which the magnetron is driven, which adjusts the RF power deliver to the linac beamline; or, alternatively, the pulse duration of the gun driver that drives an electron gun with a controlling grid, where longer pulses correlate to a greater dose per pulse and shorter pulses correlate to lower dose per pulse.

[0093] For the purpose of measuring a signal that is proportional to dose, either a scintillator crystal based detector or an ion chamber can be used. A challenge to this approach is the development of a detector with an appropriate response to the X-ray dose, and ideally with a signal level and speed that allows a pulse to pulse measurement of the X ray signal. Scintillator crystals and diode detectors are used for this purpose in the detector arrays used for X-ray imaging of cargo-they are designed for detection of the relatively lower level signals that pass through the cargo. A detector using a cadmium tungstate crystal and photodiode can be used, as is well known in the detection of gamma level X-rays. Care must be taken to keep X-ray levels low enough to minimize damage to the crystals, but high enough to provide an adequate signal for the photodiode detectors to measure with respect to electrical noise levels, as is known in the art. Detectors are available by custom order from Berkeley Nucleonics Corporation, of San Rafael California. A challenge with an ion chamber is the development of a chamber with a sufficiently fast response time and sufficiently high signal levels. One such ion chamber is the A12, which can be procured from Standard Imaging in Middleton Wisconsin.

[0094] The signal from either of these properly designed dose detectors can be used to measure dose per X-ray pulse, and this signal can be used in conjunction with a feedback circuit or computational technique for the purpose of maintaining an X-ray dose per pulse that is more stable on a pulse to pulse basis than may otherwise be achieved. The magnetron voltage is adjusted with a correction depending upon an error signal generated by the difference between the actual measured dose-per-pulse, and the desired dose-per-pulse.

[0095] Still referring to FIG. 7, an embodiment of the invention using a pulse-to-pulse dose detector can be better appreciated. It will be apparent that the basic structure is that of FIG. 1, and like elements are indicated with like numerals. As shown in FIG. 7, a pulse-to-pulse dose detector **105**, typically either scintillator-based or ion chamber-based, detects a portion of the X-rays emitted from the target, and generates a feedback signal **107** which provides an additional input to the control system **110**. The scintillator-based detector is preferred in at least some embodiments. An ion chamber can be designed to intercept and measure most of the X-ray beam by placing it close to the target. However, the ion chamber measures dose or signal in a different way than the scintillator crystal and diode combinations used in the arrays used for cargo scanning. The ion chamber measures a signal proportional to the amount of ionized gas between two conductors, but the scintillator uses a crystal that emits visible light with a relationship to the X-rays incident upon the crystal, with a photodiode and electronics being used to measure the light output of the scintillator crystal. Therefore, it is preferable in some embodiments to stabilize pulse-to-pulse dose from the linac system using a scintillator based detector.

[0096] In a typical design, a collimator (not shown) is used to generate the desired X-ray pattern for cargo scanning. In such cargo scanners, the X-ray pattern is commonly a fan beam having an included angle between 50 and 90 degrees, and a width of 2 mm to 3 mm. This line beam of MeV X-rays is used to illuminate the detector array in a scanning system, with this array made of scintillator crystals and diodes and necessary signal processing electronics. In the presently described embodiment of the invention, the pulse-to-pulse detector **105** is placed either within the linac system cabinetry itself, or near it, in a position that is never obscured by the cargo to be scanned. The detector can be placed at the edge of the X-ray fan beam, or just outside of that fan beam, in order to measure a signal that is directly proportional to that delivered in the fan beam for imaging purposes. Other locations are also possible, such as behind the electron gun, as some X-rays are emitted from the target in directions other than the fan beam direction. In an embodiment, the detector is placed in a location that provides sufficient signal for the scintillator-diode detector combination, and provides a signal that is sufficiently proportional to the dose per pulse delivered into the fan beam used for imaging.

[0097] Use of an ion chamber for measuring dose in a linac system is well known in the art. However, it is novel to use of a detector that is the same as that used in the imaging array, but here for the purpose of measuring and stabilizing the X-ray dose on a pulse-to-pulse basis. The pulse-to-pulse dose detector signals are measured by the control system, where the measured signal associated with the dose per pulse is compared against a reference value. Standard feedback techniques can be used to generate a control signal to the appropriate linac subsystem for the purpose of stabilizing the dose per pulse. In a preferred embodiment, the voltage level of the HVPS (**120**) is adjusted on a pulse-to-pulse basis in order to provide an improved stability per pulse. In another embodiment, the duration of the gun pulses is controlled in order to adjust the dose per pulse and improve the stability per pulse; in this embodiment, the duration of the pulses delivered to the grid of the electron gun is adjusted in order to adjust the dose per pulse. In a third embodiment, if a modulator is used to drive the magnetron that allows adjustment of the duration of the pulse duration to the magnetron, such as a modulator available from Scandinova Electronics or e2V, then the duration of this pulse to the magnetron can be used to adjust the dose per pulse. In all cases, benefit is provided by using a pulse-to-pulse dose detector that measures the dose per pulse in a manner that is very similar to, or exactly the same as, the manner used by the imaging array.

[0098] As discussed above in connection with FIG. 5, an embodiment of the linac system of the present invention comprises a dual grid gun driver for the purpose of providing alternating pulses of potentially different voltages and durations to the grid of the triode gun of the linac. The gun driver is used in conjunction with an electron gun in order to launch current pulses into the RF-excited linac, where the electron pulses are accelerated, and strike into a target in order to produce desired X-rays. The dual gun driver of FIG. 5 provides completely independent control of the amplitude and duration of an A pulse train and a B pulse train in a sequence of pulses that can be ABABABAB, etc., where A and B are used to represent pulses of different characteristics. In addition, it will also be appreciated, from the teachings herein, that the pulse train generated by the present invention can be configured as AAAABBBBABABABA or any arbitrary pattern that is desired for a given application.

[0099] The linac system is operated to provide controlled, accelerated electron pulses in this manner, in order to provide controlled X-ray pulses with characteristics that are desired for cargo scanning and security systems, as well as medical systems. These X-ray pulses are typically in the MeV energy range, from approximately 0.5 MeV up to 15 MeV. As discussed hereinabove, in an embodiment, the A and B pulses are at 6 MeV and 4 MeV in energy, with independently controllable doses, but can be other energies as well. In addition, in an embodiment, patterns of pulses such as ABCDABCD, where C and D represent pulses having still different characteristics from A and B, or other patterns of pulses of controllable energy and controllable dose can also be

created by the present invention.

[0100] In some embodiments, if alternating amplitudes of current are not required, the two grid driving modules shown in FIG. 5 can be reduced to one module, even if alternating or otherwise controllable pulse durations are required in a scanning application. The two grid driving modules of FIG. 5 can also be reduced to one module while still allowing alternating grid voltages if the module includes a rapidly adjustable grid top power supply. That is, if the grid top supply is designed in a manner to allow control of the grid voltage at a speed that is faster than the time between desired X-ray pulses, then the same ABABABAB functionality can be achieved, or AAAABBBBBBABABABA or any arbitrary pattern that is required. For example, an arbitrary pattern such as ABCDE may be required in an application, where A, B, C, D, E represent pulses of more than two different energies, and, in the instant case, five different energies. The resonant bandwidth of the linac imposes an upper limit on the range of different energy pulses. Other architectures are also possible, that allow grid voltage control and gun pulse duration control. In a preferred embodiment, the pulsed voltage applied to the grid of the gun is controlled between -80V and +300V with respect to the cathode voltage, and the cathode voltage is held to a voltage between -5000V and -20,000V with respect to ground. In a preferred embodiment, the durations of the grid pulses applied to the grid of the gun of the linac are controlled from between 50 nsec and 5 microseconds. These ranges are typical but do not limit the scope of the invention. Voltages such as these are used to provide peak currents at the target of as high as 200 mA, but can be controlled to as low as 1 mA or less when very low dose operation is required.

[0101] Duration control can range from a “full duration” that approaches the duration of the RF pulse (2 microseconds to 5 microseconds in preferred embodiments), down to a “minimum duration” which would provide the minimal useful integrated pulse current (such as below 100 nsec in a preferred embodiment). Such a range of control can allow rapid pulse-to-pulse adjustment of X-ray dose per pulse over a range of 5 to 1, or 10 to 1, or 20 to 1, or 50 to 1, or 100 to 1, or 200 to 1, depending on the desired effect on the X-ray pulsing.

[0102] For cargo and security scanning applications, it is desirable in some embodiments to scan a portion of the vehicle using a different dose than for other parts of the vehicle. Thus, for example, in some circumstances, it is desirable to reduce the X-ray dose during a scan of a cargo container or vehicle at one or more selected points. One reason for this is to reduce the amount of radiation required in any given scan, thereby reducing the amount of radiation to which surrounding personnel and operators will be exposed during a scan or during a collection of scans. Another reason for this is to scan part of a vehicle that may still have a driver and/or occupant or occupants in place, in which case sufficiently low dose rates are required for safety of the driver or other occupants. The linac system of the present invention, with either a dual grid gun driver or single grid gun driver (such as FIG. 11, discussed below), is controllable in a manner that enables this type of operation.

[0103] In a case where alternating energy is required, but a portion of the scan may need to be taken at a much lower dose per pulse, the pulse pattern may be abababababababababABABABABABABABA, or another arbitrary pattern, where the upper case A and B illustratively represent pulses are of alternating high and low energy, and of relatively higher dose, and the lower case “a” and “b” pulses are of the same or similar alternating high and low energy, but of relatively lower dose. FIGS. 8A-8B show the configuration of magnetron pulses **800** and **805** and gun driver pulses **810** and **815** appropriate for generating such pulses of different dose, where FIG. 8A replicates FIG. 3 as representative of a first, or “normal” dose, while FIG. 8B illustrates the different pulse characteristics that yield a second, lower, dose per pulse. The factor in doses may be of 5 to 1, or 10 to 1, or 20 to 1, or 50 to 1, or 100 to 1, or 200:1 or even more, depending on the object being scanned. For example, while the dose delivered by a ‘normal’ X-ray pulse for security scanning may be about one Gray or lower, some security scanning requires scanning the cargo area of, for example, a truck, cargo container or rail car with a dose as large as

5-10 Gray/min. In contrast, in some implementations, a driver is sitting in the cab of a vehicle to be scanned, and should not receive more than about 0.25 microSieverts in that scan, where the conversion ratio is 1 Gray (Gy)=1000 mSieverts (mSv)=1,000,000 microSieverts (μ Sv). For comparison, the average amount of radiation that an average U.S. citizen receives per year from all sources is about 3600 microSieverts. Calculating the exact ratio between Gray and microSieverts is difficult because distance from the source to the target and time of exposure are also factors, but it can readily be appreciated that the low dose pulse, suitable for scanning areas where humans are located, is typically significantly less than a normal dose pulse used for cargo scanning.

[0104] The change in dose can be accomplished by a change in gun pulse duration from one pulse to the next, or from “a” and “b” to “A” and “B”. A further change in dose can be accomplished by changing the grid voltage, which is known by those of skill in the art to change the amplitude of the current pulse that is launched from the triode electron gun into the linac. Changing only the gun pulse duration is a preferred method when the scanning application provides a benefit in that the accelerated electron energy that is associated with the “a,b” pulses and “A, B” pulses is preserved, which in an embodiment can be 6 MeV and 4 MeV. An example of the resulting pulse train is shown in FIG. 9A where the pattern of pulses shown at **900** is the “normal” dose, and the reduced dose mode is shown by the pulses at **905**. In a preferred embodiment, the dose rates at 100 pps to 400 pps during “normal dose” scanning might be between 0.1 Gy/min and 10 Gy/min, while the “reduced dose” section of a given scan might be performed at dose rates that are 2.5 $\times$, 5 $\times$, or 10 $\times$, or 20 $\times$, or 50 $\times$, or 100 $\times$ or 200 $\times$ less than the “high dose” scanning. Changing the amplitude of the gun pulse current can result in a change in the energy of the accelerated electrons in many linacs, due the effect of beam loading, which has been described. In the case of a reduced dose mode where the electron current may be 10 mA instead of 100 mA associated with a 6 MeV “normal dose” pulse, the energy associated with a lower current reduced dose pulse may be somewhat higher, such as 6.5 MeV, which is still acceptable and in some cases beneficial.

[0105] Those skilled in the art will recognize that the system of the present invention, using either a triode or a suitably fast gridded gun driver, can rapidly enable a series of reduced dose pulses immediately before, after, or during a series of normal dose pulses. Further, this technique can be advantageously applied to any linac system in a security application, so long as the linac system uses a linac device that includes a triode or suitably fast gridded gun together with a gun driver as described herein. As discussed herein, the triode or gridded gun driver can be configured to contain either two grid modulators or one grid modulator . . .

[0106] Another desirable pattern may be cccccccABABABABABA, as shown in FIG. 9B, where the series of “c” pulses **910** are of a single controllable energy and controllable dose, and the series of “A” and “B” pulses **900** is a series of pulses of alternating controllable energy and controllable dose. In a preferred embodiment, the dose of the “c” pulses is at a rate that is 5 $\times$, or 10 $\times$, or 20 $\times$, or 50 $\times$, or 100 $\times$ or 200 $\times$ less than the “high dose” A, B scanning. The energy of the “c” pulses could be the same as either “A” or “B”, or may be selected to be different.

[0107] Still another desirable pattern may be aaaaaaaaAAAAAAA, where a pulses of a single energy are used to scan, or cccccAAAAAA, where the “c” pulses of reduced dose are a different energy from the “A” pulses. FIG. 10A illustrates such a pattern of pulses, where **1000** indicates the “A” pulses and **1005** represents the “c” pulses. Either a dual grid or single grid gun driver can be used to control the linac in this manner. The dual grid can facilitate an “a” mode that differs from the “A” mode in both amplitude and in duration, whereas a single grid gun driver may be selected if only duration is to be changed on a pulse-to-pulse basis. If the “a” mode is to be the “reduced dose” pulsing mode, it can have shorter duration with respect to an “A” pulse, lower current amplitude with respect to an “A” pulse, or both. In a preferred embodiment, the linac is operated at 6 MeV, or at 4 MeV in this non-alternating mode. In another preferred embodiment, the linac is designed to provide pulses of an energy between 2 MeV and 4 MeV, for example 2.5 Mev, or, as an additional example, slightly higher than 3 Mev, such as 3.1 Mev or 3.25 Mev. with the amplitudes

and durations of the current pulses used to control the X-ray dose on a pulse-to-pulse basis.

[0108] These reduced energy levels are still sufficient to enable xray penetration of more than 200 mm of steel, which equals or exceeds the performance of systems based on Cobalt 60. For such lower energy systems, dose rate can be approximately 0.1 Gy/min at 100 pps, although such systems may be operated anywhere between 200 pps and 400 pps. It will be appreciated that a linac system that provides the same performance for cargo and security scanning systems, but without the risk factors of a radioactive material, offers substantial benefits for these applications. Such systems also have application in non-destructive testing. Operation at lower dose rates is advantageous because the zone from which personnel are excluded is smaller, and, if the dose rate required for satisfactory imagery is low enough, humans within the vehicle or object being scanned can remain in the regions being scanned. In some embodiments, it is useful to configure the dose rate generated during scanning of a first region, for example the cab of a truck where a driver is located, to be not more than 1/10, or even 1/100, of the dose rate generated during scanning of a second region, for example a cargo container carried on the back of the truck. In at least some embodiments of the present invention, the dose rate is changed by changing the dose per X-ray pulse. Dose rate can also be changed by changing the repetition rate of the pulse delivery, for example, from 400 pps to 100 pps. However, too low a pulse repetition rate will limit the resolution of the image in most scanning systems. Therefore, reducing pulse repetition rate is of only limited use in scanning applications, for example when a low dose mode is required.

[0109] When the scanning application permits use of these lower energy systems, they also offer the additional benefits of lighter weight and lower cost. At these lower energies, less shielding is required, which directly affects not only the cost, but also the mobility of the system. For example, many 6 Mev linac systems weigh over 5000 pounds, whereas a 2 Mev to 4 Mev linac system typically weighs just over 2100 pounds. The lighter system can be lawfully mounted on a two-axle vehicle, resulting in a mobile scanning vehicle with a gross vehicle weight of less than 26,000 pounds.

[0110] In an embodiment, the linac system operating in the 2 Mev to 4 Mev range is operated primarily in a pulse train of a single energy. This permits use of a simpler, single grid gun driver, further reducing cost. The linac can also be shorter than a 6 Mev linac, together with a lower power magnetron, still further reducing costs. In an embodiment, the linac is fitted with a triode gun, and the gun driver is of the grid driving type, such that the combination allows fine and rapid control of the Xray dose rate per pulse emitted from the linac system. It will be appreciated by those skilled in the art that embodiments using pulse trains of different energies within the lower range of 2 Mev to 4 Mev are also contemplated by the invention, in addition to pulse trains of a single energy.

[0111] In some cases, the speed of a change to or from a “reduced dose” mode with respect to a “normal dose” mode does not need to be on a true pulse to pulse basis, but rather can take place over several pulses, as shown in FIG. 10B, at 1010. For example, a transition to or from a “reduced dose” from/to a “normal dose” modes may still be useful if it takes place over 2 pulses or 10 pulses. In some embodiments of a gun driver, the design of the grid top power supply includes a stabilized DC power supply, and a transition between voltages may take several milliseconds, which is still a usefully fast time scale. In some embodiments, the transition from scanning a first region, with a first dose and a first energy, to scanning a second region, with a second dose and a second energy, is permitted to take 2.5, 5, 10, 25, 100, or even 250 milliseconds or more, depending upon the embodiment and the application, while in other embodiments the transition is as rapid as pulse-to-pulse. It will be appreciated that, depending upon the embodiment, the first dose and second dose can be substantially the same or different, and the first energy and second energy can be the substantially the same or different. An objective of a suitably rapid transition is to ensure that the quality of the images resulting from scanning the first region and the second region is not degraded due to the transition time required to change energies and doses. In an embodiment, the electron gun driver comprises first and second grid power supplies, each at a different voltage.

In such an embodiment, the first power supply can be configured to provide pulses for scanning the first region, and the second power supply can be configured to provide pulses for scanning the second region. As discussed elsewhere herein, the first dose can be varied from the second dose at least as the result of changes in gun driver grid voltage, gun driver pulse duration, and RF pulse power.

[0112] In still other cases, and as noted above, in some embodiments it may be desirable to change the energy as well in a “reduced dose” mode. In a preferred embodiment, the RF power can be controlled from values used in “normal dose” mode to lower values that correspond to a “reduced dose” mode that is both lower dose and lower energy. In a preferred embodiment, a capacitor charging high voltage power supply is used to charge a line type modulator to a voltage value that can be controlled to different values on a pulse to pulse basis; the subsequent voltage pulse that is applied to the RF source can therefore be controlled on a pulse to pulse basis. In an embodiment, the RF source is a magnetron. In an embodiment, the gun driver parameters are controlled synchronously with the RF parameters to provide the “reduced dose” mode.

[0113] Another method of reducing overall X-ray dose rate emitted by a system is the reduction of pulse rate. Thus, as is known in the art; a system operated at 200 pps will provide one-half the output dose rate of a system operated at 400 pps, if all other parameters are held constant. This lower repetition rate method can be used, but at some sacrifice to the resolution or speed of a given scan.

[0114] The triode gun driver, used to drive the triode electron gun of the linac, can also be configured with a single Grid Modulator **1100**, as shown in FIG. **11**. The other elements can be substantially similar, such as the High Voltage Power Supply **560** that is used to apply a DC high voltage to the electron gun cathode, an associated Filter module **555**. The DC high voltage can be approximately -20 kV in a preferred embodiment, but a range of -5 kV to -35 kV are often used with electron linacs in the MeV energy range.

[0115] The Grid Modulator is used to pulse the grid of the triode electron gun. A configuration with only one Grid Modulator can be a preferred choice for a linac that is used in primarily a single energy mode, i.e., does not require alternating back and forth between energies on a pulse to pulse basis. However, the single grid gun driver and triode gun linac, such as the embodiment described in FIG. **11**, still allows the dose rate generated by the linac to be controlled over a wide range on a pulse to pulse basis, and still allows a rapidly controllable Reduced Dose Mode that is extremely useful in applications such as cargo scanning and security scanning.

[0116] In an embodiment, the Grid Modulator **1100** includes a Bias Power Supply **1125** that is used to provide a voltage to the grid with respect to the cathode voltage for the purpose of biasing the gun in an “off” condition—the application of bias voltage is used to prevent the gun from emitting current into the linac when it is not desired.

[0117] The Grid Modulator also includes a Grid Top Power supply **1105** that is used to set a voltage that will be pulsed to the electron gun grid when pulses are desired. In a preferred embodiment, this voltage may be adjustable between -100 V and $+300$ V, and is matched to the requirements of a particular electron gun. The voltage from the Grid Top Power supply **1105** can be adjustable by the system controls, in order to control the amount of current in a pulse launched from the cathode of the triode gun into the linac structure where the electrons are subsequently accelerated. The pulse of electrons is launched via switching signals from the control system **115**, which control Switch Drive **1110**. The Switch Drive **1110** in turn controls switches **1115** and **1120** to apply to the grid either the voltage from the Grid Top Power Supply **1105** or the “off” bias voltage from the Bias Power Supply **1125**. The amplitude of the current pulse, the duration of the current pulse, and the timing of the current pulse are all used to control the dose rate from the linac on a rapid basis, and in particular can provide a Reduced Dose Mode that is very useful. The Reduced Dose Mode is useful in both single energy linac systems and alternating energy linac systems.

[0118] In another embodiment, a single Grid Modulator gun driver is used in linacs that require

alternating energy or controllable energy on a pulse to pulse basis, by configuring the Grid Top Power Supply to be a fast responding power supply instead of a DC power supply. A fast-responding Grid Top Power supply facilitates rapid change of gun pulse amplitude on a pulse-to-pulse basis, and can be used in conjunction with RF control in order to create pulse trains of varying energy, such as ABABAB, where A is 6 MeV and B is 4 MeV. Other energy configurations are also possible, such as ABCABC, or other desirable combinations between 0.5 MeV and 6 MeV, or 9 MeV or 10 MeV, or combinations in the 2 MeV to 4 MeV range, as examples.

[0119] From the foregoing, those skilled in the art will recognize that a new and novel linac-based X-ray scanning system has been disclosed, offering significant improvement in pulse-to-pulse stability for interleaved pulses of different X-ray energies, including pulse-to-pulse dose control, all on a rapidly pulsed basis. In addition, in another aspect of the invention, cool down compensation permits substantially improved stability in the dose and energy of the initial pulses after a cold start, or after a restart after an off period of indeterminate duration. Given the teachings herein, those skilled in the art will recognize numerous alternatives and equivalents that do not vary from the invention, and therefore the present invention is not to be limited by the foregoing description, but only by the appended claims.

Claims

1-27. (canceled)

28. An X-ray system for providing X-ray pulses of controlled dose per pulse comprising a linear accelerator having an electron gun, a control system for providing a first control signal to an RF source and a second control signal to an electron gun driver, the first and second control signals being independent of one another, the RF source responsive to the first control signal to supply to the linear accelerator RF pulses having a power and a duration, the electron gun driver responsive to the second control signal for providing a gun driver voltage to the electron gun for supplying to the linear accelerator pulses of controllable beam current at times substantially within the duration of a corresponding RF pulse, the RF source and the electron gun driver being controlled independently but cooperating to cause the linear accelerator to emit pulses of electrons configured to strike a target for producing X-rays of a selected energy and dose.

29. An X-ray system for providing X-ray pulses of controlled repetition rate comprising a linear accelerator having an electron gun, a control system for providing a first control signal to an RF source and a second control signal to an electron gun driver, the first and second control signals being independent of one another, the RF source responsive to the first control signal to supply to the linear accelerator RF pulses having a power and forming an envelope, the electron gun driver responsive to the second control signal for providing a gun driver voltage to the electron gun for supplying to the linear accelerator pulses of controllable beam current at times substantially within the envelope of a corresponding RF pulse, the RF source and the electron gun driver being controlled independently but cooperating to cause the linear accelerator to emit pulses of electrons configured to strike a target for producing X-rays at a selected repetition rate greater than 400 pps and less than 1,000 pps.

30. An X-ray system for providing X-ray pulses of controlled dose per pulse comprising a linear accelerator having an electron gun, a control system for providing a first control signal to an RF source comprising a modulator and an RF power source housed together and a second control signal to an electron gun driver, the first and second control signals being independent of one another, the RF source responsive to the first control signal to supply to the linear accelerator RF pulses having a power and a duration, the electron gun driver responsive to the second control signal for providing a gun driver voltage to the electron gun for supplying to the linear accelerator pulses of controllable beam current at times substantially within the duration of a corresponding RF pulse, the RF source and the electron gun driver being controlled independently but cooperating to

cause the linear accelerator to emit pulses of electrons configured to strike a target for producing X-rays of

31. The X-ray system of claim 28 wherein the duration of the RF pulses is less than 2.5 usec.

32. The X-ray system of claim 28 wherein the selected energy and dose comprises a plurality of energies and a plurality of doses.

33. The X-ray system of claim 28 wherein the selected energy and dose comprises a single energy and dose.

34. The X-ray system of claim 29 wherein the RF pulses have a duration less than 2.5 μ sec.

35. The X-ray system of claim 30 wherein the RF source further comprises a pulse transformer and the linear accelerator are cooled by a single cooling system.
